



Admixture – Types & Properties

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Admixtures - Introduction

- ❖ The term admixture as “a material other than water, aggregates, hydraulic cement, and fiber reinforcement, used as an ingredient of a cementitious mixture to modify its freshly mixed, setting, or hardened properties and that is added to the batch before or during its mixing.
- ❖ Producers use admixtures primarily to reduce the cost of concrete construction; to modify the properties of hardened concrete; to ensure the quality of concrete during mixing, transporting, placing, and curing; and to overcome certain emergencies during concrete operations.



History of Admixture

- ❖ The use of natural admixtures in concrete was a logical progression.
- ❖ Materials used as admixtures included milk and lard by the Romans; eggs during the middle ages in Europe; polished glutinous rice paste, lacquer, tung oil, blackstrap molasses,
- ❖ Extracts from elm soaked in water and boiled bananas by the Chinese; and in Mesoamerica and Peru, cactus juice and latex from rubber plants.
- ❖ The Mayans also used bark extracts and other substances as set retarders to keep stucco workable for a long period of time.



Functions of Admixtures in Concrete

- ❖ **Improve workability** of fresh concrete.
- ❖ **Reduce water requirement** for a given workability.
- ❖ **Increase strength** of concrete by reducing the water–cement ratio.
- ❖ **Accelerate setting and hardening** of concrete.
- ❖ **Delay setting time** when longer working time is required.
- ❖ **Control heat of hydration** in concrete.
- ❖ **Reduce bleeding and segregation**.
- ❖ **Improve resistance to freeze–thaw action** by introducing controlled air bubbles.
- ❖ **Reduce permeability** and improve water-tightness.



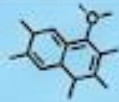
Functions of Admixtures in Concrete

- ❖ **Improve durability** against chemical and environmental attacks.
- ❖ **Reduce shrinkage and cracking.**
- ❖ **Improve pumpability** of concrete.
- ❖ **Improve slump retention** during transportation and placing.
- ❖ **Facilitate self-compacting concrete** by providing high flowability without segregation.
- ❖ **Protect reinforcement from corrosion.**
- ❖ **Improve surface finish** and appearance of concrete.
- ❖ **Reduce cement consumption** in some concrete mixes while achieving the required performance.

ADMIXTURES FOR CONCRETE



1. CHEMICAL ADMIXTURES



WATER REDUCERS
Lowers water content



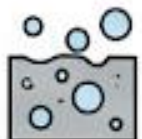
SUPERPLASTICIZERS
Enhances flowability



ACCELERATORS
Speeds up setting



RETARDERS
Delays setting



AIR ENTRAINERS
Creates air voids



**VMAs
(Viscosity Modifiers)**
Controls viscosity



2. MINERAL ADMIXTURES (SCMs)



FLY ASH
Byproduct of coal combustion



GGBS
Ground granulated slag



SILICA FUME
Fine silicon production byproduct



METAKAOLIN
Calcined kaolin clay

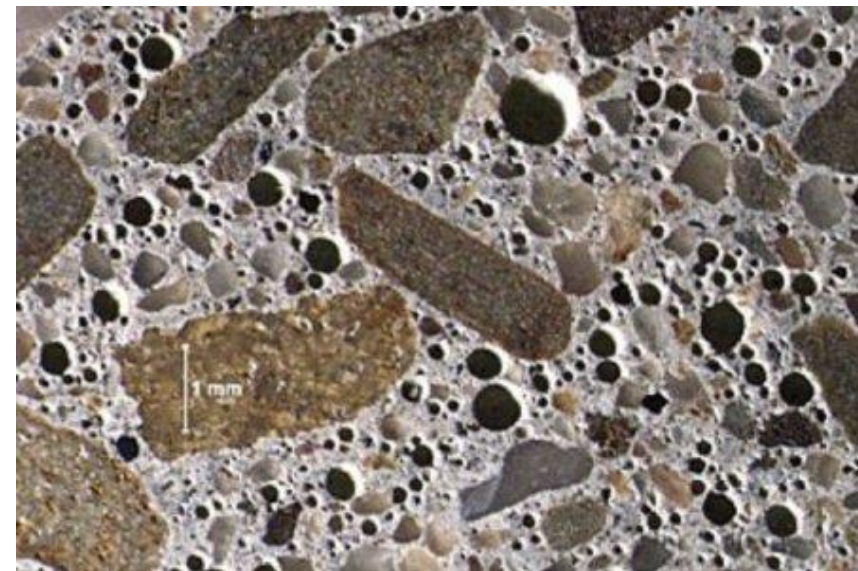


RICE HUSK ASH
Agricultural byproduct ash



Air-Entrainment Admixtures

- ❖ An air-entraining agent introduces air in the form of minute bubbles distributed uniformly throughout the cement paste.
- ❖ The main types include salts of wood resins, animal or vegetable fats and oils and sulphonated hydrocarbons.
- ❖ Entrained air is intentionally incorporated, minute spherical bubbles of size ranging from 5 microns to 80 microns distributed evenly in the entire mass of concrete.





Air-Entrainment Advantages

- ❖ Improves workability of fresh concrete.
- ❖ Reduces bleeding and segregation.
- ❖ Improves resistance to freeze–thaw cycles.
- ❖ Reduces permeability and improves durability.
- ❖ Improves resistance to surface scaling caused by freezing and de-icing salts.
- ❖ Reduces permeability of concrete.
- ❖ Improves resistance to scaling caused by free.
- ❖ Improves pumpability of concrete.
- ❖ Helps protect concrete from surface damage due to freezing and thawing.



Water Reducing (Plasticizers)

- ❖ A material, which either increases workability of freshly mixed concrete without increasing water cement ratio or maintains workability with a reduced amount of water, is termed as water reducing admixture
- ❖ As their name implies, the function of water reducing admixture is to reduce the water content of the mix, usually by 5 to 10%, sometimes (in concrete of very high workability) up-to 15%.
- ❖ Thus, the purpose of using a water reducing admixture in a concrete mix is to allow a reduction in the water cement ratio while retaining the desired workability or, alternatively, to improve its workability at a given water cement ratio.



Water Reducing (Plasticizers)



❖ The chemicals used as plasticizer (water reducing admixtures) are as follows:

1. Lignosulfonic acids, derivatives and their salts.
2. Hydroxylated carboxylic acids, their salts and derivatives.
3. Naphthalene sulphonate acid based
4. Sulfonated melamine polycondensation products



Water Reducing Advantages

- ❖ Improves workability of fresh concrete.
- ❖ Reduces bleeding and segregation.
- ❖ Improves resistance to freeze–thaw cycles.
- ❖ Reduces permeability and improves durability.
- ❖ Improves resistance to surface scaling caused by freezing and de-icing salts.
- ❖ Reduces cement consumption while maintaining the required workability and strength.
- ❖ Improves pumpability and ease of placing concrete.
- ❖ Produces denser and more uniform concrete with better surface finish.



Superplasticizer

- ❖ The admixtures capable of reducing water content by about 30% are known as super plasticizers.
- ❖ At a given water /cement ratio and water content in the mix, the dispersing action of super plasticizer increases the workability of concrete, typically by raising the slump from 75mm to 200 mm, the mix remaining cohesive.
- ❖ The resulting concrete can be placed with little or no compaction and is not subject to excessive bleeding or segregation.



Plasticizer



Superplasticizer



Superplasticizer

- There exist four main categories of super-plasticizers based on their chemical composition:
 1. Melamine formaldehyde condensates
 2. Sulfonated naphthalene formaldehyde condensates
 3. Modified lingo-sulfonates
 4. Others such as sulfonic acid esters and carbohydrate esters



Superplasticizer Advantages

- Greatly reduces water requirement in concrete.
- Improves workability and flowability of fresh concrete.
- Increases compressive strength by reducing the water–cement ratio.
- Improves durability by producing dense and less permeable concrete.
- Reduces cement consumption for a given strength and workability.
- Improves pumpability and placing of concrete.
- Useful for high-strength and self-compacting concrete (SCC).



Set-Retarding Admixture

- ❖ This type of chemical admixtures decreases the initial rate of reaction between cement and water and thereby retards the setting of concrete.
- ❖ It functions by coating the surface of C_3S (Tri calcium silicate) components, thus, delaying this reaction with the water.
- ❖ Reaction products are slow to form as such the setting and hardening of concrete are delayed reducing early compressive strengths.
- ❖ Retardation in setting time up to 8-10 hours is possible by suitable use of retarders.





Set-Retarding Admixture

The main ingredients of retarders are as follows:

- Ligno-sulphonic acids and their salts. e.g. Na, Ca or NH_4 ,
- Hydro-carboxylic acids and their salts.
- Carbohydrates including sugar.
- Inorganic salts based on flourates, phosphates, oxides, borax and magnesium salts.



Set-Retarding Advantage

- ❖ **Delays the setting time of concrete.**
- ❖ **Provides more time for transportation and placing.**
- ❖ **Improves workability in hot weather.**
- ❖ **Reduces the risk of cold joints in large concrete pours.**
- ❖ **Helps control rapid setting caused by high temperature.**
- ❖ **Facilitates finishing of concrete over a longer period.**
- ❖ **Useful for long-distance transportation of ready-mix concrete.**
- ❖ **Helps maintain uniformity and workability during prolonged placing operations.**



Accelerating Admixture

- ❖ These admixtures when added to concrete, mortar or grout increases the rate of hydration of hydraulic cement, shortens the time of set, accelerates the hardening or development of strength of concrete/ mortar.
- ❖ These admixtures function by interaction with C_3S (Tri- calcium silicate) component of the cement thus increasing the reaction between cement and water.
- ❖ Many substances are Known to Act as accelerators for concrete. They include Alkali Hydroxides, Silicates, Fluoro-Silicates, Organic Compounds, Calcium Formates, Calcium Nitrates, Calcium Thio Sulphate, Aluminium Chlorides, Potassium Carbonates, Sodium Chlorides & Calcium Chlorides.



Accelerating Admixture Advantages

- ❖ Shortens the setting time of cement and therefore increases the rate of gain of strength.
- ❖ Enables earlier release from precast moulds thus speeding production.
- ❖ Reduces segregation and increase density and compressive strength.
- ❖ Cures concrete faster and therefore uniform curing in winter and summer can be achieved.
- ❖ Reduces water requirements, bleeding, shrinkage and time required for initial set.



Fly Ash

- ❖ Fly ash is finely divided residue resulting from the combustion of powdered coal and transported by the flue gases and collected by electrostatic precipitator.
- ❖ Fly ash is the most widely used pozzolanic material all over the world.
- ❖ The importance and use of fly ash in concrete has grown so much that it has almost become a common ingredient in concrete, particularly for making high strength and high performance concrete.
- ❖ The utilisation of fly ash as a supplementary cementitious material. High volume fly ash concrete is a subject of current interest all over the world.



Fly Ash

- ❖ The use of fly ash as concrete admixture not only extends technical advantages to the properties of concrete but also contributes to the environmental pollution control.
- ❖ There are two ways that the fly ash can be used: one way is to inter grind certain percentage of fly ash with cement clinker at the factory to produce Portland pozzolana cement (PPC) and the second way is to use the fly ash as an admixture at the time of making concrete at the site of work.
- ❖ One of the important characteristics of fly ash is the spherical form of the particles. This shape of particle improves the flowability and reduces the water demand. The suitability of fly ash could be decided by finding the dry density of fully compacted sample.



Effect of Fly Ash on Fresh Concrete

- ❖ Use of right quality fly ash, results in reduction of water demand for desired slump.
- ❖ With the reduction of unit water content, bleeding and drying shrinkage will also be reduced.
- ❖ Since fly ash is not highly reactive, the heat of hydration can be reduced through replacement of part of the cement with fly ash.
- ❖ The reduction of temperature rise for 30% substitution of fly ash.



Effect of Fly Ash on Hardened Concrete

- ❖ Fly ash, when used in concrete, contributes to the strength of concrete due to its pozzolanic reactivity. However, since the pozzolanic reaction proceeds slowly, the initial strength of fly ash concrete tends to be lower than that of concrete without fly ash.
- ❖ Due to continued pozzolanic reactivity concrete develops greater strength at later age, which may exceed that of the concrete without fly ash.
- ❖ Fly ash concrete should be cured for longer period. In this sense, fly ash concrete used in under water structures such as dams will derive full benefits of attaining improved long term strength and water-tightness.



Silica Fume

- ❖ Silica fume, also referred to as micro silica or condensed silica fume, is another material that is used as an artificial pozzolanic admixture.
- ❖ It is a product resulting from reduction of high purity quartz with coal in an electric arc furnace in the manufacture of silicon or ferrosilicon alloy.
- ❖ Silica fume rises as an oxidised vapour. It cools, condenses and is collected in cloth bags.
- ❖ It is further processed to remove impurities and to control particle size. Condensed silica fume is essentially silicon dioxide (more than 90%) in non crystalline form. Since it is an airborne material like fly ash, it has spherical shape. Influence on Fresh Concrete.



Effect of Silica Fume on Fresh Concrete

- ❖ **Reduces workability** due to its very fine particles and high surface area.
- ❖ **Increases water demand** in the concrete mix.
- ❖ **Reduces bleeding** by retaining water within the mix.
- ❖ **Improves cohesiveness** and reduces segregation.
- ❖ **Increases viscosity**, so superplasticizer is often required to maintain workability.
- ❖ Reduces slump if the mix proportions are not properly adjusted.
- ❖ Improves uniformity of the fresh concrete by reducing segregation.
- ❖ May make finishing more difficult because of increased stickiness and viscosity.



Effect of Silica Fume on Hardened Concrete

- ❖ **Increases compressive strength** due to the pozzolanic reaction and dense microstructure.
- ❖ **Increases tensile and flexural strength** of concrete.
- ❖ **Reduces permeability** by filling the pores and capillary voids.
- ❖ **Improves durability** against chloride penetration and chemical attack.
- ❖ **Reduces water absorption** due to the denser concrete structure.
- ❖ **Improves bond strength** between cement paste and aggregates.
- ❖ **Reduces carbonation-related deterioration** when the concrete is properly designed and cured.



Ground Granulated Blast Furnace Slag

- GGBFS is a by-product of the iron and steel manufacturing process in a blast furnace.
- It has a high specific gravity, generally about 2.8–3.0.
- Its fineness is usually slightly higher than that of cement.
- Different types of blast furnace slag include:
 - Air-cooled slag
 - Expanded or foamed slag
 - Granulated slag
 - Pelletized slag
- Granulated slag is commonly used as a mineral admixture in concrete.
- Granulated slag is rapidly quenched with water and then ground into a fine powder.
- The finely ground material is known as Ground Granulated Blast Furnace Slag (GGBFS).



Ground Granulated Blast Furnace Slag

- GGBFS has both cementitious and pozzolanic properties.
- An activator is required for the hydration of GGBFS.
- It has little effect on workability because its fineness is close to that of cement.
- Replacement of cement with GGBFS slows the early strength development of concrete.
- GGBFS generally improves the ultimate strength of concrete.
- It improves the durability of concrete.
- It increases resistance to chemical attack and corrosion.
- It is particularly useful in marine structures exposed to aggressive environments.
- Concrete containing GGBFS may show a higher rate of carbonation compared with ordinary Portland cement concrete.



Ground Granulated Blast Furnace Slag





Performance of GGBS in Concrete

- ❖ The replacement of cement with GGBS will reduce the unit water content necessary to obtain the same slump.
- ❖ This reduction of unit water content will be more pronounced with increase in slag content and also on the fineness of slag. This is because of the surface configuration and particle shape of slag being different than cement particle.
- ❖ In addition, water used for mixing is not immediately lost, as the surface hydration of slag is slightly slower than that of cement
- ❖ Reduced heat of hydration
- ❖ Refinement of pore structures
- ❖ Reduced permeabilities to the external agencies
- ❖ Increased resistance to chemical attack.



Rice Husk Ash

- ❖ Rice husk ash is obtained by burning rice husk in a controlled manner without causing environmental pollution.
- ❖ When properly burnt it has high SiO_2 content and can be used as a concrete admixture. Rice husk ash exhibits high pozzolanic characteristics and contributes to high strength and high impermeability of concrete.
- ❖ India produces about 122 million ton of paddy every year. Each ton of paddy produces about 40 kg of RHA. There is a good potential to make use of RHA as a valuable pozzolanic material to give almost the same properties as that of microsilica.





Surkhi

- ❖ Surkhi, was the commonest pozzolanic materials used in India.
- ❖ Surkhi is an artificial pozzolana made by powdering bricks or burnt clay balls. In some major works, for large scale production of surkhi, clay balls are specially burnt for this purpose and then powdered.
- ❖ By its nature, it is a very complex material differing widely in its qualities and performances. Being derived from soil, its characteristics are greatly influenced by the constituent mineral composition of soil, degree of burning and fineness of grinding.
- ❖ Because of the complexity of problem there has been much confusion on account of contradictory results obtained by various research workers.

